Ganji Sujeeth Madhav

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PROFESSIONAL SUMMARY

B.Tech Computer Science (AI & ML) graduate with hands-on experience building systems end-to-end - from embedded hardware and inverse kinematics to full-stack AI-driven applications. Comfortable with SQL, data structures & algorithms, and Python; enjoys solving problems with no existing reference implementation to follow.

EDUCATION

B.Tech, Computer Science and Engineering (Artificial Intelligence & Machine Learning)

Holy Mary Institute of Technology and Science, Ghatkesar, Telangana | Expected 2026 | GPA: 8.5/10.0

SKILLS

- Programming & Data: Python, SQL (MySQL), Data Structures & Algorithms, HTML, JavaScript
- AI/ML: TensorFlow, OpenCV, LangChain, Vector Embeddings (HuggingFace), AprilTags
- Systems & Tools: Flask, Raspberry Pi, ESP32, Godot/GDScript, Git/GitHub, Browser extension development
- Soft Skills: Problem Solving, Communication, Teamwork, Quick Learning

PROJECTS

Dynamic AI-Driven RPG Game Framework — *2025 (Graduation Project)*

- Built a proof-of-concept game engine where story, maps, NPCs, quests, and items are generated dynamically by AI rather than hand-authored, using a schema-constrained approach: an LLM (via LangChain + Groq) generates structured JSON specs which Godot deterministically parses and executes.
- Designed a procedural map generator combining Perlin-noise biome generation with a tunneling and constrained random-walk dungeon carving algorithm, and a flood-fill-based system that tiers dungeon room loot by distance from the entry point.
- Implemented dependency-ordered content generation for NPCs (auto-generating required quests and items before the NPC itself) and branching, quest-state-aware dialogue trees.
- Used HuggingFace embeddings in a vector database to give the game long-term memory of player actions and past lore, enabling quest generation that references a player's actual history.

Tech: *Godot, GDScript, Python, Flask, LangChain, Groq, HuggingFace Embeddings*

Vision & Gamepad-Controlled 3-DOF Robotic Arm — *2024*

- Designed and built a 3-DOF robotic arm from 3D-printed parts and servo motors, controlled via Raspberry Pi 3B+.
- Manually derived the arm's inverse kinematics using the DH parameter method and wrote a custom Python library to convert target coordinates into servo angles, with no existing reference implementation available.
- Built two independent control interfaces: real-time hand/index-finger tracking via computer vision, and joystick/gamepad control.

Tech: *Python, OpenCV, Raspberry Pi, Inverse Kinematics*

Marks Entry Automation — *2026*

- Built a browser extension and a Python server that automate student FA/SA mark entry on a government school's online portal.
- The extension extracts the school metadata from the portal and The server generates a per-subject Excel sheet keyed by student ID, converts the filled sheet to JSON, and the extension uses it to enter marks, attendance, and absent markers.
- Deployed the server on Render's free tier so it runs without a local machine and Handled portal-specific rules, including the Telugu default language, class-based subject lists, and the four-slot mark structure that determines grades.

Tech: *Python, JavaScript, AprilTags, Deployment on Render, Browser extension development*

ESP32 One-Time Password Access Device — *2023-2024*

- Designed a physical authentication device (ESP32 + button + LCD) that generates a one-time password with a 10-second validity window, displayed only on the device itself.
- Built a secure access-check flow where entering the correct password unlocks protected data on a connected system.